Structural Algebras of Admissibility, Execution, and Observation in Discrete Mechanics

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The Axiomatic Cascade of Change and the Continuability Constraint

The structural architecture of discrete mechanics is constructed by running two separate derivations toward one another: a downward cascade from the single constraint that all things must change, and an upward operational ladder resting on physical transformation, electronics, and digital computation. The downward path begins at the primordial ground of genuine nothingness, recognized as the "Wash"—an undifferentiated condition wherein all signal cancels locally, meaning total connection reads locally as no connection . To establish a reference within this ground requires a direction, which introduces structural differentiation .

The transition from stasis to dynamic execution is governed by the C_1 continuability constraint, which mandates that every executable state must possess a distinct legal successor, thereby forbidding the existence of terminal fixed points . This requirement presupposes a structural gap—designated as the C_0 or "comma" identity—across which difference can be serialized and measured . The physical instantiation of the C_0 comma is demonstrated by a boundary walk on a cylinder versus a Möbius strip . For a cylinder of size $N = 12$ with no comma (offset 0), a boundary walk visits only a single face, resulting in a period of 12 (equivalent to $1N$), which yields no reachable distinction . On a Möbius strip containing the comma (offset 1), the walk serially reads both faces, resulting in a period of 24 ($2N$), which serializes the superposition into a readable temporal distinction .

Boundary Walk Configuration	Faces Visited	Period Length	Temporal Character
Cylinder (No Comma, Offset 0)	``	12 ($1N$)	Stasis; no distinction
Möbius Strip (Comma, Offset 1)	``	24 ($2N$)	Serialized superposition

When the continuability constraint is coupled with a binary difference operator Δ, the resulting combinatorial dynamics are strictly constrained . If novelty is formalized as the monotone accumulation of visited states ($N(t) = |states\ visited|$), a greedy novelty-seeking walk on n-bit states produces a maximal-period de Bruijn cycle without importing external feedback shift register theory . This has been computationally verified across widths $n \in \{4,5,6,7,8\}$.

Alternatively, expressing the difference as a linear recurrence over $GF(2)$ yields an m-sequence . However, every linear m-sequence possesses a unique C_1 violation: the all-zero state at the origin behaves as an absorbing fixed point . To compel execution, the feedback bit of the origin must flip, which breaks bijectivity at its target . To restore bijectivity, the feedback of exactly one other state must also flip . An exhaustive search over all possible feedback modifications proves that the minimal repair is of size two, unique at every tested width, and consists precisely of the origin and its algebraic opposite, $hi = 1 \ll (n-1)$. The repaired dynamics yield a single continuous orbit of length 2^n .

The linear cycle structure and its patched de Bruijn counterpart are documented across widths $n \in \{4,5,6,7\}$:

Lane Width (n)	Linear Cycles (with Violation)	Patched Cycles (with Repair)	Uniqueness of Minimal Repair
$n = 4$	(Fixed point at)	`` (de Bruijn cycle)	Minimal size 2; repair: ``
$n = 5$	(Fixed point at)	`` (de Bruijn cycle)	Minimal size 2; repair: ``
$n = 6$	(Fixed point at)	`` (de Bruijn cycle)	Minimal size 2; repair: ``
$n = 7$	(Fixed point at)	`` (de Bruijn cycle)	Minimal size 2; repair: ``

Under this patched formulation, a uniform transition measure is not an assumed axiom of symmetry, but rather an emergent exhaust of the forced repair . An earlier hypothesis that uniform transition measure is an independent axiom is refuted; uniform in-degree is one realization of equal transition measure, and the uniform measure falls out as a structural consequence of $C_1 + \Delta$.

Analyzing these dynamics at the vertex level introduces an artificial indeterminacy, as vertices with multiple incoming paths make forward predictions history-dependent . This is resolved by lifting the state from the vertex to the edge, defined as the ordered pair of the previous and current positions . On the edge space, evolution is a single-valued, strict bijection .

The first-order difference represents velocity, and the discrete second difference Δ^2 represents acceleration [PULL]. On the 6-bit de Bruijn sequence, Δ^2 was measured and shown to be multi-valued on vertices (occupying 32 distinct sets over 32 nodes) but single-valued on edges (occupying 0 ambiguous sets over 64 edges) . This establishes that mass-state and acceleration are functionally native to the same enriched edge state, representing successive rungs on a single jet tower .

The Restoration Price and Traveler Forgetting

The downward path requires that a difference be maintainable under transformation, while the upward path details the physical cost of doing so . Digital computation does not preserve analog signals; it rebuilds boundaries . Simulating a signal travelling down a chain of 100 stages under attenuation and noise reveals that a passive chain degrades to pure chance, while a restoring chain utilizing regenerating gain holds the signal open .

Stage Count	Passive Chain Error (No Gain)	Restoring Chain Error (Gain 3)
3	0.0267	0.0001
5	0.1548	0.0000
10	0.4093	0.0000
20	0.4960	0.0001
50	0.4950	0.0005
100	0.5014 (Coin Flip)	0.0014 (Restored)

The convergence of these paths yields a forced law: exactness requires restoration, restoration requires gain above unity, gain requires a source, and the source is the price .

A record's survival is also governed by forgetting-by-dilution . Unlike non-injective erasure, dilution does not ontologically destroy early ledger levels . Rather, their share of the expanding address space shrinks geometrically, and they stop constraining new territory . In the cascade, this dilution decay is measured quantitatively as a deficit that drops from 0.53 down to 0.26, and further down to 0.09 .

A parallel boundary governs traveler death . An early hypothesis that traveler removal is caused by constancy or period-length limits is retracted; a traveler modeled as a change-sensitive high-pass filter survived period-two inputs at 67.4%, while an adaptive predictor of length 32 flattened under a period-97 sinusoid, which requires only two taps of memory .

The actual variable is environmental complexity relative to model capacity . Driving a 24-tap predictor with random blocks of length L repeated indefinitely shows a sharp threshold: when $L \leq 24$, the environment is fully compressed, prediction error drops to zero, and response amplitude flattens to 0.0% . When $L > 24$, the environment exceeds model capacity, and the traveler remains alive, registering high response amplitude .

Repeated Block Length (L)	Response Surviving	Verdict
4	0.0%	Flat (Dead)
8	0.0%	Flat (Dead)
16	0.0%	Flat (Dead)
20	0.0%	Flat (Dead)
24 (Capacity Limit)	0.0%	Flat (Dead)
28	98.4%	Alive
40	98.6%	Alive
80	99.7%	Alive
Never Repeats	98.5%	Alive

The Commutator and the Continuity Dichotomy

The primary measurement of interaction within the transformation space is the commutator:

$$= ABA^{-1}B^{-1}$$

If A and B commute, they leave no residue; otherwise, they leave a geometric residue . An early hypothesis that mass represents displacement resistance is retracted . Because every basic generator acts as an avalanche mixer, the displacement channel saturates at the thermal floor ($W/2$ bits) with an indistinguishable spread-to-mean ratio of 0.008 across bodies .

The correct settlement channel identifies mass as the base-two logarithm of the commutator's mathematical order:

$$O(\lambda) = \log_2 \text{ord}(\lambda) \quad [1, 1]$$

Because permutation order is independent of the start state, mass calculated in this channel is intrinsic and center-independent .

A highly persistent early hypothesis that opacity (the lack of fixed points) is identical to inertia (high order) is retracted . A commutator that is a fixed-point-free involution has an order of exactly 2 (settlement mass of 1 bit), yet possesses zero fixed points . It is simultaneously nearly weightless and completely opaque, demonstrating that mass and transparency are independent projections of the same commutator .

The structural separation of fixed points (F) and order (O) is formalized by their incompatible continuity classes under geometric perturbation . Fixed points are Lipschitz bounded:

$$||Fix(\sigma P)| - |Fix(P)|| \leq |\text{supp}\ \sigma| \quad [1, 1]$$

A minimal support-3 surgery can alter the fixed-point count by at most 3, regardless of N . In contrast, the order has no modulus of continuity . At $N = 2^k$, a transposition of support 2 can split a single N-cycle into coprime lengths $s = N/2 - 1$ and $N - s$, yielding an order of $s(N-s) = N^2/4 - 1$. The ratio of new to old order is: $\frac{\text{ord}(\tau P)}{\text{ord}(P)} = \frac{N}{4} - \frac{1}{N}$ At $N = 64$ and $s = 31$, this ratio is exactly 16, rising from 64 to 1023 . The order is a sum over prime-power maxima via the exponent map:

$$O(\lambda) = \sum_q e_q(\lambda) \log_2 q \quad \text{where} \quad e_q(\lambda) = \max_i v_q(L_i) \quad [1, 1]$$

Because a maximum lacks locality, a single newly injected prime alters the global order irrespective of where it occurs .

A lane's mechanical spectrum possesses a single resonance indexed by the existence of a push equal to its own inverse—an involution ($a = W/2$), present if and only if W is even . An early hypothesis that resonance is indexed by the divisor lattice is retracted; a width of $W = 15$ with divisors 3 and 5 remains flat under sweep .

The commutator of a body with the half-swap involution (S) expands to a product of two involutions, each fixing $\sqrt{N}$ points . Under the Fragmentation Theorem, a product of two involutions fixing k points fragments the space into k cycles of mean length N/k . Setting $k = \sqrt{N}$ forces $\sqrt{N}$ cycles of length $\sqrt{N}$. The involution-line coefficient:

$$C = \frac{\log_2(\text{order})}{\sqrt{N}}$$

was hypothesized to be a universal constant, but is retracted; C shifts from 1.45 to 1.71 depending on ensemble and mixing depth .

Iterating commutators of near-commuters drives support down to a fundamental algebraic floor of 3 points, which is the minimal support of the alternating group . This represents the irreducible quantum of change .

Structure is anisotropic mass [PULL]. Building a body along a commuting axis preserves fixed points and keeps it transparent to its own axis while paying full price on others, whereas depth accumulated across non-commuting axes thermalizes the body to a Poisson(1) baseline of $E[|Fix|] = 1.00$. This is verified under a matched cycle-type control experiment :

Depth (d)	Commuting Graph E[|Fix|]	Broken Graph E[|Fix|]
2	8.77	3.35
4	6.14	1.35
8	6.54	0.91
16	7.58	1.01
32	7.32	0.90
64	7.68	0.95
128	5.05	1.07 (Poisson Floor)

The Combinatorial Dynamics of Cycle Surgery

Permutation surgery on the cycle spectrum is governed by the topological split and merge rules of Theorem 1 . Let $P \in S_\Omega$ and let transposition $\tau = (x, y)$ act on the left :

- **Split:** If x and y share a cycle C of length L, and $y = P^s(x)$ for $1 \leq s \leq L - 1$, then C splits into two cycles of lengths s and $L - s$.
- **Merge:** If x and y occupy distinct cycles C_1 and C_2 of lengths L_1 and L_2, they merge into a single cycle of length $L_1 + L_2$.

A single transposition changes the cycle count by exactly $\Delta K = \pm 1$. Because a minimal-support surgery factors as $(a, b, c) = (a, c)(a, b)$, its effect is computed by applying the transposition rule twice sequentially . The change in cycle count ΔK is determined by the topological span $\sigma \in \{1,2,3\}$:

- **Span 3 ($\sigma = 3$):** All three points occupy distinct cycles. Two merges occur, yielding $\Delta K = -2$.
- **Span 2 ($\sigma = 2$):** Two points share a cycle and the third is distinct. One split and one merge occur, yielding $\Delta K = 0$.
- **Span 1 ($\sigma = 1$):** All three points share a single cycle. The first split is followed by either a merge or a second split, yielding $\Delta K \in \{0, +2\}$ depending on the cyclic order of a, b, c .

This topological span rule was verified over 3000 random surgeries: 1765 span-3 cases produced $\Delta K = -2$, 1140 span-2 cases produced $\Delta K = 0$, and span-1 cases split 48: 47 between $\Delta K \in \{0, +2\}$. The merge-length sum law held in all 231 qualifying trials .

Under the Compression Law, the successor spectrum is computable in $O(L_{max})$ time from purely localized index-position data $((i_a, \pi_a), (i_b, \pi_b), (i_c, \pi_c))$, bypassing full permutation arithmetic . In a stress test of

1800 trials across widths $W \in \{8,10,12\}$ for both pure and rich bases, the predicted spectrum matched the executed spectrum in 1800 out of 1800 cases .

The thermodynamic driver of this system is prime injection via the failure of prime-powers to close under addition . If a permutation P has all cycle lengths equal to powers of a single prime π, merging two cycles of lengths π^a and π^b ($a > b$) yields a merged length of:

$$\pi^a + \pi^b = \pi^b(1 + \pi^{a-b}) \quad [1, 1]$$

Since any prime divisor of $1 + \pi^{a-b}$ is coprime to π, the merge strictly enlarges the prime support . Applying identical surgeries to pure spectra ($W = 12, |S| = 1$) generated mean mass gains of $+6.20$ and $+6.35$ bits, while rich random permutations ($|S| \in \{8,9\}$) eroded by -2.56 and -0.01 bits . Coherence and order-fragility are the same property .

Under uniform surgery selection, the cycle partition is a mathematically sufficient statistic for the spectrum process . In an empirical validation at $W = 10$, the total variation distance between the transition distribution of a pure multiplication and its randomly relabeled conjugate was 0.2442, matching the same-source control distance of 0.2398 . The trajectory is high-entropy: from a single partition, 4000 surgeries produced 972 distinct successor spectra, yielding an empirical conditional entropy lower bound of $H(\lambda_{t+1} \mid \lambda_t) \geq 7.49$ bits .

The observable layer contains no internal compositional structure, acting as a flat coordinate system . Evaluating two-port joint attractors against the linear segment interpolating single-port attractors:

Joint Port Pairing	**Port Type Class**	**Normalized Segment Distance (d/|seg|)**
$O \downarrow$ with $F \uparrow$	Arithmetic × Local Geometric	0.11
$O \downarrow$ with $K \downarrow$	Arithmetic × Global Geometric	0.29
$p \downarrow$ with $M \downarrow$	Arithmetic × Scale	0.20
$O \downarrow$ with $p \downarrow$	Arithmetic × Arithmetic (Control)	0.35
$F \uparrow$ with $K \downarrow$	Geometric × Geometric (Control)	0.25

The joint attractors fall within the noise floor of the linear interpolating segments . The hypothesis that ports compose as an algebra is retracted; joint feedback merely forces a weighted compromise between singular optima .

At $W = 12$, the expected order change $D(p) = E = G - \Lambda$ decomposes into a flat gain G from newly injected primes and a loss Λ that scales linearly with the current prime count p :

Prime Count (p)	Sample Size (n)	Expected Gain (E[G])	Expected Loss (E[Λ])	Drift (D(p))
5	91	11.15	6.46	+5.58
6	220	11.38	9.67	+2.52
7	394	12.09	10.73	+2.10
8	483	11.50	12.08	+0.18
9	423	11.16	12.81	−0.92
10	235	9.75	14.19	−3.71
11	98	8.37	15.21	−6.02
12	29	8.83	15.71	−6.29

The stable macroscopic equilibrium (p^*) scales linearly with the lane width W :

$$p^*(W) \approx 0.94W - 2.8 \quad [1, 1]$$

An earlier reading of $p^* \approx 6.2$ as a universal constant is retracted, having been a width-specific measurement at $W = 10$.

Lane Width (W)	State Space Size (N)	Expected Gain (E[G])	Exponent Erosion Coefficient (c)	Predicted p*	Measured p*
8	256	5.67	0.870	4.11	4.71
10	1024	8.02	1.108	5.71	6.22
12	4096	11.35	0.974	7.54	8.12
14	16384	14.24	0.583	8.74	10.32

Conjugate Pairs, Channel Transport, and the Gauge Invariance

Physical domains transport change as conjugate pairs whose product is power . Measuring a physical drivetrain crossing three domains—a battery source, a motor (gyrator), a gearbox, and a lead screw (transformer)—reveals that power is conserved across stages, while the impedance ratio is local .

Drivetrain Stage	**Effort (e)**	**Flow (f)**	**Product (e·f)**	**Impedance Ratio (e/f)**
Battery Source	24.0000	3.5000	84.00000000 W	6.8571
Motor (Gyrator, $r = 0.42$)	1.4700	57.1429	84.00000000 W	0.0257
Gearbox (Transformer, $m = 7.5$)	11.0250	7.6190	84.00000000 W	1.4470
Lead Screw (Transformer, $m = 0.08$)	0.8820	95.2381	84.00000000 W	0.0093

Across the 7.5 ratio gearbox, the impedance ratio scales by 56.250000, which is exactly $m^2 = 7.5^2$.

An early hypothesis that the commutator's conjugate pair is represented by its trace and norm is retracted . Because the trace of any commutator is identically zero, their product is always zero and carries no information . The correct decomposition splits any operator product into its symmetric and antisymmetric halves : $$AB = \frac{1}{2}{A, B} + \frac{1}{2}

$$The symmetric half carries the bilinear invariant pairing tr(AB) (analogous to physical power), while the antisymmetric half$$

is the residue . Under a transformer channel, tr(AB) is invariant (drift 8.9×10^{-1}), $$ is invariant (drift exactly 0.00), and the ratio of their magnitudes scales as m^2 (56.250000 for $m = 7.5$), matching the physical drivetrain .

An ideal channel has a determinant magnitude of 1, and a lossy channel has determinant η . Channel efficiency η and erasure bits per pass conform to:

$$\text{bits per pass} = -\log_2|\det M| = -\log_2 \eta \quad [1, 1]$$

Measured Channel	**Efficiency (η=|detM|)**	**Predicted Erasure**	**Measured Erasure**
Ideal Transformer ($m = 7.5$)	1.000000	0.000000	0.000000

Measured Channel	**Efficiency (η=|detM|)**	**Predicted Erasure**	**Measured Erasure**
Ideal Gyrator ($r = 0.42$)	1.000000	0.000000	0.000000
Writer Projection (Edge to Vertex)	0.500000	1.000000	1.000000 (Exhaustive)
Crush: Keep 8 of 16 bits	0.003906	8.000000	8.000000 (Exhaustive)
Crush: Keep 4 of 16 bits	0.000244	12.000000	12.000000
Lossy Transducer ($\eta = 0.85$)	0.850000	0.234465	0.234465
Lossy Transducer ($\eta = 0.50$)	0.500000	1.000000	1.000000

The rank-two limit on this law is closed . Skew-symmetry annihilates every odd coefficient of the characteristic polynomial, making the true invariant count $\lfloor n/2 \rfloor$, which represents a multiport bond graph carrying parallel bonds (verified for $n = 2$ to 12) .

Continuously, the bit loss rate is carried by the symmetric half of the generator:

$$\frac{dB}{dt} = -\frac{\mathrm{tr}(M)}{\ln 2} \quad [1, 1]$$

The skew half's flow has a determinant of 1.0000000000 at all times tested . A generator with zero trace preserves volume and loses no bits (Liouville's theorem); a generator with a negative trace contracts volume and erases bits (Landauer's principle) .

The hyperbolic form is the unique quadratic form preserved by the measured channel group, whose stabilizer in GL_2 is exactly $O(1,1)$ (drift 0.00) . The null directions of this form correspond to the carrier axes, indicating that effort and flow are the light-cone coordinates of the transported pairing . Rotating these axes to the definite form $e^2 + f^2$ (stabilizer $O(2)$) preserves the definite form (drift 2.7×10^{-18}) but breaks the hyperbolic form (0.703) . The abstract pairing space is two-dimensional, but the physically realized group selects the hyperbolic form . The invariant ring is a polynomial ring on a single degree-2 generator: $\mathbb{R}[e \cdot f]$. Odd degrees carry zero invariants, and even degrees carry exactly $(e \cdot f)^{d/2}$. The 1+1 Lorentz group, spacetime geometry, or confinement are noted as structural analogies [PULL].

The Graph Representation of Software and Congruence Quotients

Finite closed deterministic computations are classified by their transition graphs up to graph isomorphism, with symbol systems acting as coordinate charts . The Gray-counter toggle and binary increment counters share a functional graph of shape $(8,)$, and the dictionary between them was constructed by search as:

$$\sigma(x) = x \oplus (x \gg 1) \quad [1]$$

This is the Gray code, constructed from the transition topology without importing arithmetic .

On the 6502 CPU, the operation field aaa carries its operation whole (3.0000 bits, zero leakage), and the mode field bbb carries its mode whole (3.0000 bits), but the hex display's fixed 4 | 4 nibble grid tears the mode across the boundary, with 1 bit high and 2 bits low . The functional boundary is pinned by execution; the nibble is a coordinate laid at an angle over weight spaces .

Open Mealy machines extend this classification to input-indexed, output-labeled transition graphs . Running a search over all 96 triples of simultaneous relabelings between a serial subtractor and a serial adder returns exactly four dictionaries that represent the two classical subtraction-via-addition identities . The canonical family maps:

$\sigma = \text{swap(borrow, carry)},\ \pi: (a, b) \mapsto (a, \neg b),\ \rho = \text{identity}$ [1]This is the two's-complement arithmetic identity [1]. The second family maps:$\sigma = \text{identity},\ \rho = \text{flip},\ \pi: (a, b) \mapsto (b, \neg a)$ [1]

This represents the one's-complement of the minuend identity, $\neg(a - b - w) = \neg a + b + w$. A negative control against a D-latch returned exactly 0 of 96 dictionaries, providing a formal equivalence boundary .

Implementation is a congruence quotient, meaning a partition whose classes map to single classes, valid over all states . Putnam-style triviality—where any physical system implements any program by mapping a single trajectory—fails this condition: alternately labeling one trajectory of a random map claims a 2-cycle, and the all-states lawfulness test returns False .

The Distributive Lattice and the Two Bills

Reality is composed of three separate, interacting algebras:

- **MAY (Constraints)** → $\mathcal{L}$**:** A complete distributive lattice .
- **DOES (Execution)** → $\mathcal{M}$**:** An execution monoid .
- **SEEN (Observation)** → Q**:** An observational quotient .

The operations of these layers satisfy different mathematical laws, meaning they cannot be collapsed ($\mathcal{L} \neq \mathcal{M} \neq Q$) . An FPGA behaves as a physical laboratory version of this separation . The physical hardware fabric CLBs and switch matrices form a pre-existing "legal connection space" constraint lattice ($\mathcal{L}_{FPGA}$), while clocked transitions propagate state via transition algebra composition ($\mathcal{M}$), and the physical pin signals represent the observational quotient (Q) . Neighborhood and spatial lattice structures emerge naturally because physical interaction costs impose local constraints .

On open machines, serial composition fails to descend to isomorphism classes if the interface is forgotten; renaming the first machine's output alphabet by a permutation produces three distinct composite classes from six relabelings . Transporting the identification along the interface collapses the result to a single invariant class . The interface is not a label; it is part of the morphism, represented as (M, I) .

Rename operations show that implementation details and contract are separated by the boundary :

Relabeling Applied to Composite	Pairs Tested	Composition Survives
Internal State Names (Implementation)	3,600	100.0%
Wire Message Alphabet (Contract)	900	27.2%
— Identity Permutation Only	150	100.0% (150/150)
— Single Transposition	150	51,24,12 of 150
— Three-Cycle	150	5,3 of 150

An early composition experiment on total protocols is retracted as vacuous, as a contract that forbids nothing is not a contract; partiality is what defines an interface .

Interface cost is a filter, not a meter . Driving composites to 400 steps reveals that deaths cease, with cumulative failure saturating at 0.9687 by step 100 and 0.9690 by step 400 . The surviving runs fall into a boundary-compatible sublanguage .

Filters compose by conditioning, not multiplication; applying the same filter twice guarantees survival ($P(F_2 \mid F_1) = 1.0000$) .

Filter F1	Filter F2	P(F1)	P(F2)	P(F1 ∧F2)	P(F1)P(F2)	P(F2|F1)	Joint Bits vs Sum
(a,c,b)	(a,c,b)	0.0297	0.0297	0.0297	0.0009	1.0000	5.0710 vs 10.1419
(a,c,b)	(b,a,c)	0.0306	0.0296	0.0115	0.0009	0.3755	6.4422 vs 10.1062
(a,c,b)	(c,b,a)	0.0310	0.0301	0.0127	0.0009	0.4113	6.2934 vs 10.0645
(b,c,a)	(c,a,b)	0.0125	0.0129	0.0107	0.0002	0.8600	6.5395 vs 12.6012

Because the lattice composition is idempotent, any additive valuation is trivial:

$$f(A) = f(A \wedge A) = 2f(A) \Longrightarrow f(A) = 0$$ [1, 1]

This explains why the interface layer resists the accumulating $-\log_2$ ledger .

Execution acts on constraints by precomposition (pullback), which preserves the lattice structure with zero violations . Forward image $T(\cdot)$ fails meet in 733 of 3000 trials, failing exactly at non-injective transitions . Ahead-of-time verification is sound if and only if the admissible set is closed under execution:

$$T(A) \subseteq A \quad [1, 1]$$

This condition separated 387 closed cases (100.0% sound) from 3613 open cases (0.0% sound), establishing subject reduction as a basic legality condition for compilation .

The constraint and execution operators partition into at least four classes based on their algebraic signatures :

Operator	**Idempotent**	**Shrinking**	**Expanding**	**Monotone**	**Meets**	**Joins**	**Class**
ASSERT (Meet with Fixed Set)	Yes	Yes	No	Yes	Yes	Yes	Lattice Homomorphism
ν (Largest Invariant Subset)	Yes	Yes	No	Yes	Yes	No	Interior Operator
μ (Orbit Closure)	Yes	No	Yes	Yes	No	Yes	Closure Operator
Forward Image $T(\cdot)$	No	No	No	Yes	No	Yes	Execution
Parity Gate	Yes	Yes	No	No	No	No	Pathological

Because the compilation operator ν is an interior operator, it fails join, giving the falsifiable prediction that a disjunctive property can be statically checkable when neither disjunct is . This was verified with the cycle-straddling sets $A = (\nu(A) = \emptyset)$ and $B = (\nu(B) = B)$, where $\nu(A \cup B) =\supset \nu(A) \cup \nu(B)$.

An interface is defined by its basis (P, R), where P partitions the source (with an inadmissible cell) and R realizes the mapping to the target . Every interface role is generated by (P, R) :

- constrain$(I) \Leftrightarrow |Adm| < |S|$.
- translate$(I) \Leftrightarrow \exists x \in Adm$ with $R(x) \neq x$.
- preserve$(I) \Leftrightarrow |R^{-1}(y)| = 1$ for every y in the image .

This basis generates two distinct bills :

- **Admission Price:** $A(I) = -\log_2$, which is idempotent ($A(I \circ I) = A(I)$) and sub-additive .
- **Crossing Price:** $X(I) = H(X \mid Y)$, which is additive and paid per crossing .

A validating interface that is injective on what it admits has $A = 1.0000$ and $X = 0.0000$ bits, providing a free crossing . The biconditional is exact: preserve holds if and only if $X(I) = 0$.

Under composition, crossing prices telescope: $X(g \circ f) = X(f) + X(g)$ where Y carries the pushforward measure of the source through f . In 8 random chains, additivity held exactly (e.g., $1.272783 + 0.844361 = 2.117144$ and $1.647783 + 1.154025 = 2.801808$) . A naive shortcut utilizing the average fibre size:

$$X = \sum_y \frac{|R^{-1}(y)|}{n} \log_2 |R^{-1}(y)|$$

is restricted to uniform sources, and failed in every trial under pushforward conditions .

The crossing price equals the selector cost of its inverse relation:

$$X(f) = H(X \mid Y) = E_Y[\log_2 |f^{-1}(y)|] = \text{selector}(f^{-1}) \quad [1]$$

This was verified to machine precision across five maps :

Map Configuration (f)	**Target Image Size**	**H(X|Y)**	**Selector Cost of f−1**	**Match**
Identity Map	$16/16$	0.000000	0.000000	True
Halving Map ($x//2$)	$8/16$	1.000000	1.000000	True
Quartering Map ($x//4$)	$4/16$	2.000000	2.000000	True
Reduction Modulo 3 ($x \pmod 3$)	$3/16$	2.420566	2.420566	True

Map Configuration (f)	Target Image Size	H(X|Y)	Selector Cost of f−1	Match
Random Map	$7/16$	0.750000	0.750000	True

State-admissibility does not determine dynamics . Enumerating all 24 permutations of 4 states and grouping them by their lattices of invariant subsets yields 15 distinct structures, with the largest class containing 6 distinct dynamics that share a single invariant-set lattice . The projection from transition to state structure costs 0.979574 bits in expectation, with the 6-member fibre losing $\log_2 6 = 2.5850$ bits .

The apparent rescue that transition-level or hyperedge-level admissibility separates all dynamics is definitional, since the admissible transition set of a deterministic system is the graph of its map . Below a system's event arity, every description is a projection charging $-\log_2$ of the fibre . Under species-level unary projections of 4000 sampled reaction networks, the pairwise view loses 0.9908 bits, and the species view loses 8.6978 bits .

The reversibility of a map and its state-level legibility trade off along a strictly monotone curve :

True Crossing Price (H(X|Y))	Associated Fibre Partitions	Map Count	Mean Bits Lost by State Summary
0.00000	$(1,1,1,1)$	24	0.9796
0.50000	$(2,1,1)$	144	0.7642
1.00000	$(2,2)$	36	0.6667
1.18872	$(3,1)$	48	0.5000
2.00000	$(4,\emptyset)$	4	0.0000

The Functorial Split and the Entropy Seam

The projections from the parent transition object split into functorial invariants and level-relative measurements . The cycle spectrum transports functorially along quotient morphisms . In 1496 checks across the congruences of an 8-cycle, Random Map A, and Random Map B, quotient cycle lengths divided parent cycle lengths with zero violations .

In contrast, the step-count and image-collapse tolls do not transport functorially . A near-bijection on 8 states with parent toll 0.1926 bits has a lawful two-class quotient whose toll is 1.0000 bits, representing a $5.19\times$ amplification .

Transition Map	Lawful Congruences	Spectrum Divisibility Violations	Parent Toll (Bits)	Maximum Quotient Toll (Bits)	Toll Monotone?
8-Cycle (Permutation)	4	$0/32$	0.0000	0.0000	Yes
Random Map A	315	$0/1260$	0.6781	2.0000	No
Random Map B	88	$0/176$	0.6781	2.0000	No
Near-Bijection ($\|im\| = 7$)	4	$0/28$	0.1926	1.0000	No

The toll is conjugation-invariant but level-relative, appearing only where the resolution level drops . Restricting then observing differs from observing then restricting in 49.1% of 2000 trials, with the coarse route admitting more states than it should ($\{0,2,3\}$ vs $\{0,3\}$) . This pipeline non-commutation: MAY → DOES → SEENis the entropy seam . The spectrum is what remains when the projection commutes; the toll is what appears when it does not . The loss is not inside the object; it is created in the map . Entropy is defined as the obstruction to a measurement commuting with coarse-graining:

$$S_\pi := \text{Failure of measurement to commute with projection } \pi \quad [1]$$

This is proven as a finite structural obstruction within the transition model, with its thermodynamic identification held as a structural analogy [PULL].

Unmeasured Experimental Setups

The remaining operational and algebraic structures generate specific proposed experiments whose empirical outcomes are not yet provided.

Feedback Composition on Open Machines

Serial and parallel compositions have been analyzed, but the algebraic structure of feedback composition on open machines remains unmeasured.

Setup: Couple the output of Mealy machine M_1 back to its own input via a feedback interface I_f.

Constraints: Interface I_f must be a validating, non-total, non-identity contract.

Metric to be measured: The count of distinct closed state trajectories and the resulting cycle spectrum.

Conclusion: ____

Disjunctive Checkability in Static Analyzers

The interior operator ν predicts that a disjunctive property can be statically checkable when neither disjunct is checkable on its own, which can be evaluated against existing static analysis tools.

Setup: Run a static analysis of a program containing disjunctive loop invariants $A \vee B$ over non-prefix-closed sets.

$$\text{Constraints: Invariants } A \text{ and } B \text{ must individually fail to satisfy the closed-form preservation condition } T(X) \subseteq X.}$$

Metric to be measured: Pass/fail validation rate of the joint invariant $A \vee B$ versus individual invariants.

Conclusion: ____

Hardware Realization of Idempotent Constraint Instructions

It has been hypothesized that physical processors implement specific instructions mimicking the algebraic properties of the constraint lattice.

Setup: Profile instruction execution times on real processor architectures executing repeated invariant check routines.

Constraints: Invariant checks must act as a projection onto a fixed set (ASSERT style).

Metric to be measured: The energy cost and cycle count of executing the projection twice sequentially versus once.

Conclusion: ____

The Dynamical Equivalence of Ledger Depth and Time

The formulation that time is ledger depth depends on a direct structural correspondence between the depth of the algebraic record and physical execution rounds.

Setup: Trace an executing de Bruijn sequence program and record its cumulative bit expansion rate.

$$\text{Constraints: Program must execute under a constant, non-zero continuous loss rate } tr(M) = -0.165515.}$$

Metric to be measured: Linear correlation between the discrete execution steps and the geometric address space expansion.

Conclusion: ____